

Sai Amulya Yellapragada

linkedin.com/in/saiamulya | yellapragadasaiamulya@gmail.com | +1 240-733-5371 | <https://github.com/Sai-Amulya>

EDUCATION

Master of Science in Software Engineering

January 2024 – December 2025

Arizona State University

Bachelor of Engineering, Electronics and Communications Engineering

June 2016 – September 2020

Stanley College of Engineering & Technology for Women, Hyderabad, India

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, HTML, CSS, Spring Boot, React Native, SQL, C++, C#, Unity

Tools & Technologies: Docker, Gradle, Maven, Git, GitHub, Jira, ANTLR, JUnit, Jest, Selenium, Expo

Software Development Methodologies: Agile, Scrum, DevOps, CI/CD

PROFESSIONAL EXPERIENCE

Software Engineer, Accenture, India

December 2020 – December 2023

- Designed and developed RESTful APIs using Java and Spring Boot for product catalog and order management, optimizing performance with asynchronous processing, query tuning, and caching, improving response times by 25%
- Built and maintained microservices with Spring Boot, improving modularity, scalability, and development speed
- Migrated legacy monolithic services to AWS-based microservices architecture using Spring Boot and Docker, improving backend system performance
- Developed Python automation scripts to ingest bulk user data, process product and order records for analytics, and generate sales and inventory reports, improving data accuracy and reducing manual effort by 80%
- Enhanced search capabilities with Elasticsearch, boosting query speed and relevance by 30%

PROJECTS

Floe Programming language | Python, ANTLR

- Co-developed Floe, a cross-platform imperative programming language enhanced with logic programming features, supporting comprehensive data types, control structures, and user-defined functions
- Designed and implemented a custom lexical analyzer, parser (using ANTLR), and runtime environment in Python 3 to enable seamless language processing and execution

Sales Forecasting | Python

- Built and evaluated machine learning models for Walmart sales forecasting using Kaggle's historical dataset, applying feature engineering and training models such as Random Forest, Decision Tree, XGBoost, and LightGBM
- Achieved accuracy of 97.38%, demonstrating strong predictive performance for weekly sales trends through model tuning and validation

Scrum Board Simulator | Java, Spring Boot, Gradle, MySQL

- Collaborated within a team of five to enhance a Scrum simulation platform with user story management, sprint controls, and role-based access per SRS; followed Agile practices over two sprints, completing 25 tasks and delivering 85 business value points, tracked via Taiga burndown charts
- Developed features using Java Spring Boot with modular architecture, GoF design patterns, and Gradle; used GitHub for version control with structured branching and code reviews, and created unit tests to ensure reliability and maintain test coverage

WORK EXPERIENCE

Course Grader, Arizona State University, Seasonal

August 2024 – Present

- Graded 200+ student assignments, exams for a course on Integrated Business Processes and enabling technologies, evaluating student understanding of ERP data flow, transaction processing, and database management
- Assessed 60+ projects, assignments and case studies for undergraduate and graduate level statistics courses, supporting practical application of statistical analysis, quality management, and Six Sigma methodologies

Community Advocate- ASU LIFT Initiative, Arizona State University

August 2024 – July 2025

- Led peer mentoring programs for first-year graduates, fostering belonging and leadership, enhancing academic success

Graduate Service Assistant, Arizona State University

February 2024 – November 2024

- Developed Python-based predictive models achieving 96% accuracy in forecasting Li-ion battery State of Health (SoH) for proactive EV battery management
- Analyzed Lithium-ion battery data using MATLAB & Origin to uncover performance patterns and optimize battery management systems